

Proposed Revision of the Assumptions

1 Introduction

2 Related Literature

3 Model and Target Parameters

This section presents the dynamic panel model, defines the low-rank coefficient matrices and their fitted values, and introduces the linear target parameters considered in the paper.

3.1 Model Specification

We observe a balanced panel of outcomes and covariates for units $i = 1, \dots, N$ and periods $t = 1, \dots, T$. The model is

$$y_{it} = \sum_{\ell=1}^P a_{0,it,\ell} y_{i,t-\ell} + \sum_{k=1}^K x_{it,k} \beta_{0,it,k} + h_{0,it} + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T. \quad (3.1)$$

Here, $y_{i,t-\ell}$ is the ℓ th lag of the outcome, $x_{it,k}$ is the k th observed covariate, and u_{it} is an idiosyncratic error. We write $x_{it} = (x_{it,1}, \dots, x_{it,K})'$ when referring to the covariates jointly. The term $h_{0,it}$ is an unobserved interactive-effects component that captures additive common, unit, time, and interactive heterogeneity and is treated as a nuisance component. The autoregressive coefficients $a_{0,it,\ell}$ and slope coefficients $\beta_{0,it,k}$ may vary across both units and time. The initial values $y_{i,0}, \dots, y_{i,1-P}$ are observed, while the numbers of lags P and covariates K are fixed and known.

The model contains no separate intercept, and x_{it} does not include a constant with an unrestricted unit- and time-varying coefficient. Such a coefficient matrix would enter additively and would not generally be separately identified from the interactive-effects matrix. Additive heterogeneity is therefore absorbed by $\mathbf{H}_0$.

For each lag ℓ , let $\mathbf{A}_0^{(\ell)} = (a_{0,it,\ell})_{N \times T}$ denote the autoregressive coefficient matrix. For each covariate k , let $\mathbf{B}_0^{(k)} = (\beta_{0,it,k})_{N \times T}$ denote the slope coefficient matrix, and let $\mathbf{H}_0 = (h_{0,it})_{N \times T}$

denote the interactive-effects matrix. Each matrix has its own fixed rank:

$$r_{A,\ell} = \text{rank}\left(\mathbf{A}_0^{(\ell)}\right), \quad \ell = 1, \dots, P, \quad (3.2)$$

$$r_{B,k} = \text{rank}\left(\mathbf{B}_0^{(k)}\right), \quad k = 1, \dots, K, \quad (3.3)$$

$$r_H = \text{rank}(\mathbf{H}_0). \quad (3.4)$$

When the ranks are referenced jointly, we collect them in $\mathbf{r}_0 = (r_{A,1}, \dots, r_{A,P}, r_{B,1}, \dots, r_{B,K}, r_H)$. The ranks may differ across matrices and may equal zero.

Low rank allows the coefficients to vary across units and time while keeping the effective dimension of each matrix small relative to NT . For example, if $\mathbf{A}_0^{(1)}$ has rank one, then $a_{0,it,1} = \lambda_i f_t$ for a unit-specific loading λ_i and a common time component f_t . Higher ranks permit several such components, and no common rank or shared factor structure is imposed across the coefficient matrices.

3.2 Matrix representation

We collect a generic candidate set of coefficient matrices and the interactive-effects matrix in

$$\Theta = \left(\mathbf{A}^{(1)}, \dots, \mathbf{A}^{(P)}, \mathbf{B}^{(1)}, \dots, \mathbf{B}^{(K)}, \mathbf{H}\right) \in \mathbb{H}_{NT} := (\mathbb{R}^{N \times T})^{P+K+1}. \quad (3.5)$$

For two such collections, define

$$\langle \Theta_1, \Theta_2 \rangle_{\mathbb{H}} = \sum_{\ell=1}^P \left\langle \mathbf{A}_1^{(\ell)}, \mathbf{A}_2^{(\ell)} \right\rangle + \sum_{k=1}^K \left\langle \mathbf{B}_1^{(k)}, \mathbf{B}_2^{(k)} \right\rangle + \langle \mathbf{H}_1, \mathbf{H}_2 \rangle, \quad (3.6)$$

where $\langle \mathbf{C}_1, \mathbf{C}_2 \rangle = \sum_{i=1}^N \sum_{t=1}^T C_{1,it} C_{2,it}$ is the entrywise inner product.

For observation (i, t) , the fitted value generated by Θ is

$$\mathcal{Y}_{it}(\Theta) = \sum_{\ell=1}^P A_{it}^{(\ell)} y_{i,t-\ell} + \sum_{k=1}^K B_{it}^{(k)} x_{it,k} + H_{it}. \quad (3.7)$$

The argument of $\mathcal{Y}_{it}(\cdot)$ identifies the collection being evaluated. Thus, $\mathcal{Y}_{it}(\Theta_0)$ is the true systematic component, $\mathcal{Y}_{it}(\hat{\Theta})$ is a fitted value, and $\mathcal{Y}_{it}(\Delta)$ is the fitted-value change generated by a collection of coefficient changes Δ having the same structure as Θ . Model (3.1) can therefore be written as

$$y_{it} = \mathcal{Y}_{it}(\Theta_0) + u_{it}. \quad (3.8)$$

3.3 Parameters of Interest

The estimator is defined for deterministic linear summaries of the autoregressive and slope coefficient matrices. The interactive-effects matrix is a nuisance component and is not a target in the analysis

below. For a target index ν , write

$$\phi_\nu(\Theta_0) = \langle D_\nu, \Theta_0 \rangle_{\mathbb{H}}, \quad (3.9)$$

where D_ν has the same component structure as Θ_0 and has a zero component for H_0 .

Examples include $\beta_{0,it,k}$ or $a_{0,it,\ell}$ for a specified unit and time period. For a unit group G_N and a time set R_T , a group-time mean uses weights $(|G_N||R_T|)^{-1}$ on the corresponding coefficient entries. A full cross-sectional mean at time t sets $R_T = \{t\}$ and $G_N = \{1, \dots, N\}$. For disjoint groups $G_{1,N}$ and $G_{2,N}$, the between-group slope contrast at time t is

$$\frac{1}{|G_{1,N}|} \sum_{i \in G_{1,N}} \beta_{0,it,k} - \frac{1}{|G_{2,N}|} \sum_{i \in G_{2,N}} \beta_{0,it,k}. \quad (3.10)$$

Because P is fixed, the lag-sum persistence index $\sum_{\ell=1}^P a_{0,it,\ell}$ and its group averages are also linear targets. This index is not an impulse response or long-run multiplier.

There is an important distinction between defining an estimand and proving its limiting distribution. Theorem ?? applies only when the target satisfies Assumption 6. The global first-stage result in Theorem ?? does not by itself verify that condition for an individual entry, a one-period group mean, or their contrasts. Appendix ?? records the target directions and the additional localized control these examples require. Nonlinear response paths and long-run multipliers remain outside the formal theory.

4 The Cross-Fitted Debiased Estimator

5 Assumptions

This section states the six maintained statistical assumptions: the data-generating process, low-rank signal, identification and first-stage regularity, and target-specific inference regularity. The additional conditions used only for the appendix rank-consistency theorem are stated with that theorem in Appendix ?. Asymptotics are taken jointly as $N, T \rightarrow \infty$, while P, K , and the true matrix ranks remain fixed. Let $\mathbb{P}_0$ denote the probability law generated by the true parameter collection Θ_0 , and let $\mathbb{E}_0$ denote expectation under $\mathbb{P}_0$. The true parameter matrices and initial states are treated as fixed. Let $\mathcal{F}_{t-1}$ denote the information generated by these fixed objects and the observed panel history through time $t - 1$. The fold assignment is independent of the panel and is not conditioned out in $\mathbb{P}_0$ or $\mathbb{E}_0$.

For unit i at time t , let $C_{0,it}$ be the companion matrix formed from $a_{0,it,1}, \dots, a_{0,it,P}$.

Assumption 1 (Stable dynamics and stochastic predetermined regressors).

(i) There exist constants $C_* < \infty$ and $\rho_* \in (0, 1)$ such that

$$\sup_{\substack{i \leq N, h \geq 1 \\ t+h \leq T}} \|\mathbf{C}_{0,i,t+h} \cdots \mathbf{C}_{0,i,t+1}\|_{\text{op}} \leq C_* \rho_*^h. \quad (5.1)$$

(ii) The predetermined covariates admit a possibly nonstationary innovation representation. For $h \geq 1$, let $x_{it,k}^{(h)}$ be obtained by replacing the primitive innovation at time $t - h$ with an independent copy while holding all other primitive innovations fixed. Uniformly over i, t, k , and h ,

$$\left\{ \mathbb{E}_0 \left| x_{it,k} - x_{it,k}^{(h)} \right|^{8+\eta} \right\}^{1/(8+\eta)} \leq C_* \rho_*^h. \quad (5.2)$$

(iii) For units i and j and $h \geq 0$, let $x_{i,t+h,k}^{\langle j,t \rangle}$ be obtained by replacing the unit-specific primitive innovations of unit j at time t with an independent copy, while holding all aggregate and common innovations fixed. There is a nonincreasing function $\omega_x(r) \downarrow 0$ such that

$$\left\{ \mathbb{E}_0 \left| x_{i,t+h,k} - x_{i,t+h,k}^{\langle j,t \rangle} \right|^{8+\eta} \right\}^{1/(8+\eta)} \leq C_* \rho_*^h \omega_x(d_N(i, j)). \quad (5.3)$$

The initial states of the outcome and covariate recursions are uniformly bounded.

The second clause requires the effect of a past innovation on a current covariate to decline geometrically with its time distance. The third requires the effect of a unit-specific innovation to decline with both time and cross-sectional distance. Assumption 5(ii)–(iii) specifies explicit spatial-decay and buffer-growth rates. These rates are also used to control the observation-specific coupling in Proposition ??.

Assumption 2 (Sequential exogeneity and temporal innovation structure). *For every i and t ,*

$$\mathbb{E}_0[u_{it} \mid x_{it}, \mathcal{F}_{t-1}] = 0. \quad (5.4)$$

The primitive innovation vectors driving the covariates and outcomes are independent across t .

Assumption 2 permits current and future covariates to depend on past idiosyncratic errors. It does not impose strict exogeneity. The stronger conditional centering needed only for an entry's locally important periods is stated in Assumption 6(ii), rather than imposed on every target.

Assumption 3 (Moments and weak cross-sectional dependence). *There exist constants $\eta > 0$, $0 < c_\sigma < C_\sigma < \infty$, and $C < \infty$ such that, almost surely and uniformly in i, t, k ,*

$$\begin{aligned} c_\sigma &\leq \mathbb{E}_0[u_{it}^2 \mid x_{it}, \mathcal{F}_{t-1}] \leq C_\sigma, & \mathbb{E}_0[|u_{it}|^{8+\eta} \mid x_{it}, \mathcal{F}_{t-1}] &\leq C, \\ \mathbb{E}_0|x_{it,k}|^{8+\eta} &\leq C. \end{aligned} \quad (5.5)$$

The same conditional variance and $8 + \eta$ moment bounds continue to hold after additionally conditioning on the current aggregate and common innovations entering the covariates.

Conditional on $\mathcal{F}_{t-1}$ and those current aggregate and common innovations, the field $\{(u_{it}, x_{it}) : i \leq N\}$ is strongly mixing. More precisely, for any disjoint unit sets A and B , the conditional mixing coefficient between the sigma-fields generated by the observations in A and B is bounded by $\alpha\{d_N(A, B)\}$, where $d_N(A, B) = \min_{i \in A, j \in B} d_N(i, j)$ and $\alpha(r) \downarrow 0$.

The $8 + \eta$ bounds allow non-Gaussian and moderately heavy-tailed errors and regressors. The mixing condition concerns how dependence weakens with cross-sectional distance; it is not a sub-Gaussian tail assumption. Assumption 5(ii) supplies the explicit spatial geometry and decay rates used by the concentration and spatial central-limit arguments.

Assumption 4 (Pervasive and well-spread low-rank components). *For each ℓ with $r_{A,\ell} > 0$, write $\mathbf{A}_0^{(\ell)} = \mathbf{U}_{A,\ell} \boldsymbol{\Sigma}_{A,\ell} \mathbf{V}_{A,\ell}'$. For each k with $r_{B,k} > 0$, write $\mathbf{B}_0^{(k)} = \mathbf{U}_{B,k} \boldsymbol{\Sigma}_{B,k} \mathbf{V}_{B,k}'$, and, when $r_H > 0$, write $\mathbf{H}_0 = \mathbf{U}_H \boldsymbol{\Sigma}_H \mathbf{V}_H'$. There exist constants $0 < c_{\text{sig}} \leq C_{\text{sig}} < \infty$ and $\mu_{\text{inc}} < \infty$, uniform over all positive-rank matrices, such that*

$$c_{\text{sig}} \sqrt{NT} \leq \sigma_{r_{A,\ell}}(\mathbf{A}_0^{(\ell)}) \leq \sigma_1(\mathbf{A}_0^{(\ell)}) \leq C_{\text{sig}} \sqrt{NT}, \quad (5.6)$$

$$\max_i \|\mathbf{e}_i' \mathbf{U}_{A,\ell}\|_2^2 \leq \frac{\mu_{\text{inc}} r_{A,\ell}}{N}, \quad \max_t \|\mathbf{e}_t' \mathbf{V}_{A,\ell}\|_2^2 \leq \frac{\mu_{\text{inc}} r_{A,\ell}}{T}, \quad (5.7)$$

$$c_{\text{sig}} \sqrt{NT} \leq \sigma_{r_{B,k}}(\mathbf{B}_0^{(k)}) \leq \sigma_1(\mathbf{B}_0^{(k)}) \leq C_{\text{sig}} \sqrt{NT}, \quad (5.8)$$

$$\max_i \|\mathbf{e}_i' \mathbf{U}_{B,k}\|_2^2 \leq \frac{\mu_{\text{inc}} r_{B,k}}{N}, \quad \max_t \|\mathbf{e}_t' \mathbf{V}_{B,k}\|_2^2 \leq \frac{\mu_{\text{inc}} r_{B,k}}{T}. \quad (5.9)$$

When $r_H > 0$, the same inequalities hold with $(\mathbf{B}_0^{(k)}, r_{B,k}, \mathbf{U}_{B,k}, \mathbf{V}_{B,k})$ replaced by $(\mathbf{H}_0, r_H, \mathbf{U}_H, \mathbf{V}_H)$.

The ranks are matrix specific and may differ across lags, covariates, and the interactive-effects matrix. The singular-value bounds impose pervasive signals, while the singular-vector bounds prevent concentration on a small number of units or time periods.

Let $m_{NT} = \min(N, T)$. For each component matrix M , let $r_{M,0}$ be its true rank and $\bar{r}_M$ its cap in $\mathcal{R}_{\text{max}}$. Define

$$\mathcal{D}_{\text{max}} = \{\boldsymbol{\Delta} : \text{rank}(\boldsymbol{\Delta}_M) \leq \bar{r}_M + r_{M,0} \text{ for every } M\}, \quad \mathcal{D}_{\text{max}}(1) = \{\boldsymbol{\Delta} \in \mathcal{D}_{\text{max}} : \|\boldsymbol{\Delta}\|_{\mathbb{H}} = 1\}. \quad (5.10)$$

Let $\mathcal{T}_0$ denote the product tangent space at $\boldsymbol{\Theta}_0$; rank-zero components contribute the zero tangent space.

Let b_{NT} be a deterministic truncation threshold used only in the proofs. It satisfies

$$\frac{(NT)^{1/(8+\eta)}}{b_{NT}} \rightarrow 0, \quad \frac{b_{NT}^4 \log(NT J_{NT})}{m_{NT}} \rightarrow 0. \quad (5.11)$$

Define

$$\zeta_{NT} = b_{NT}^2 \log(NT J_{NT})(N^{-1} + T^{-1}). \quad (5.12)$$

The sequence b_{NT} is an analytical truncation envelope used to state the rates implied by the $8 + \eta$ moments. It is not an additional tail assumption and is not estimated from the data.

Assumption 5 (Identification, panel structure, and sample splitting).

(i) *Population identification.* There exist constants $0 < c_S < C_S < \infty$ such that

$$c_S \|\Delta\|_{\mathbb{H}}^2 \leq \mathbb{E}_0 \sum_{i=1}^N \sum_{t=1}^T \mathcal{Y}_{it}(\Delta)^2 \leq C_S \|\Delta\|_{\mathbb{H}}^2 \quad (5.13)$$

for every difference between Θ_0 and a parameter in a fixed rank class used by the estimator or rank selector, with component ranks bounded by $\bar{r}_M$.

(ii) *Panel growth and spatial geometry.* Both panel dimensions diverge, and N/T is bounded away from zero and infinity. For the entrywise spatial limit and spatial-kernel results, the unit locations lie in a fixed-dimensional Euclidean space, distinct locations are uniformly separated, and d_N is their Euclidean distance. For constants $C_d, c_d > 0$ and $d_s < \infty$, uniformly in N and $r \geq 0$,

$$v_N(r) \leq C_d(1+r)^{d_s}, \quad \omega_x(r) + \alpha(r)^{\eta/(8+\eta)} \leq C_d \exp(-c_d r). \quad (5.14)$$

The rank caps $\bar{r}_M$, as well as P and K , remain fixed.

(iii) *Fold and buffer growth.* The number of folds J_{NT} , the forward buffer q_T , and the spatial buffer r_N are deterministic researcher-controlled sequences satisfying

$$J_{NT}, q_T, r_N \rightarrow \infty, \quad q_T + r_N = O\{\log(NT)\}. \quad (5.15)$$

They also satisfy

$$\frac{(q_T + 1)v_N(r_N)}{J_{NT}} + \frac{J_{NT}(N + T) \log(NT J_{NT})}{NT} \rightarrow 0, \quad (5.16)$$

$$b_{NT}^2 J_{NT} (N^{-1} + T^{-1}) \log\{(N + T)J_{NT}\} \rightarrow 0, \quad (5.17)$$

$$\sqrt{NT} \left[\rho_*^{q_T} + \sum_{r > r_N} v_N(r)^3 \left\{ \omega_x(r) + \alpha(r)^{\eta/(8+\eta)} \right\} \right] \rightarrow 0. \quad (5.18)$$

Part (i) is the low-rank analogue of no perfect collinearity: different admissible coefficient matrices must generate sufficiently different fitted values. Part (ii) requires a balanced panel, controlled growth of local spatial neighborhoods, and rapid decay of temporal and cross-sectional dependence. The exponential inequality concerns dependence across time and space; the marginal tails remain governed by the $8 + \eta$ moment framework in Assumption 3. In part (iii), q_T removes future dynamic feedback and r_N removes nearby contemporaneous dependence. The constant ρ_* is the dynamic-decay constant already defined in Assumption 1; it is not an additional tuning parameter. Proposition ?? derives fold balance, sample retention, and the first-stage concentration bounds from these primitive conditions.

For a scalar target $\phi_\nu(\Theta_0) = \langle D_\nu, \Theta_0 \rangle_{\mathbb{H}}$, let $\mathcal{P}_0$ be the projection onto the true product tangent space and define $W_\nu = \mathcal{P}_0 D_\nu$. Assumption 5(i) makes the population sample-second-moment form

coercive on $\mathcal{T}_0$. Hence there is a unique population Riesz direction $\bar{\mathbf{q}}_\nu \in \mathcal{T}_0$ satisfying

$$\mathbb{E}_0 \sum_{i=1}^N \sum_{t=1}^T \mathcal{Y}_{it}(\boldsymbol{\Delta}) \mathcal{Y}_{it}(\bar{\mathbf{q}}_\nu) = \langle \mathbf{W}_\nu, \boldsymbol{\Delta} \rangle_{\mathbb{H}} \quad \text{for every } \boldsymbol{\Delta} \in \mathcal{T}_0. \quad (5.19)$$

Equation (5.19) therefore defines $\bar{\mathbf{q}}_\nu$; it is not an additional identifying assumption.

Define

$$\tilde{\Psi}_{\nu,it} = \mathcal{Y}_{it}(\bar{\mathbf{q}}_\nu), \quad Z_{\nu,it} = \tilde{\Psi}_{\nu,it} u_{it}, \quad \xi_{\nu,t} = \sum_{i=1}^N Z_{\nu,it}. \quad (5.20)$$

The variable $Z_{\nu,it}$ is the observation-level influence contribution: it is the first-order contribution of observation (i, t) to the target estimation error. Let

$$v_{\nu,t} = \mathbb{E}_0(\xi_{\nu,t}^2 \mid \mathcal{F}_{t-1}), \quad V_{\nu,NT} = \sum_{t=1}^T v_{\nu,t}, \quad s_{\nu,NT}^2 = \mathbb{E}_0 V_{\nu,NT}. \quad (5.21)$$

Only a fixed number of components of $\mathbf{D}_\nu$ are nonzero, and each has fixed rank. Write

$$\|\mathbf{D}_\nu\|_{*,1} = \sum_{\ell=1}^P \|\mathbf{D}_\nu^{A,\ell}\|_* + \sum_{k=1}^K \|\mathbf{D}_\nu^{B,k}\|_*. \quad (5.22)$$

This class contains individual entries, group means, between-group contrasts, full means, and fixed lag sums.

Let

$$\mathcal{D}_{\nu,NT} = \{d_{\nu,1} < \dots < d_{\nu,K_\nu}\} \quad (5.23)$$

be the deterministic set of target periods allowed to contribute a nonvanishing share of the target variance, where K_ν is uniformly bounded. Put $d_{\nu,0} = 0$ and $d_{\nu,K_\nu+1} = T+1$, and define the variance contributed by the diffuse periods between two successive target periods as

$$V_{\nu,k}^{\text{seg}} = \sum_{t=d_{\nu,k}+1}^{d_{\nu,k+1}-1} v_{\nu,t}, \quad k = 0, \dots, K_\nu. \quad (5.24)$$

For a fully diffuse target, $K_\nu = 0$. For a fixed entry, $\mathcal{D}_{\nu,NT}$ contains its target time.

For $t \in \mathcal{D}_{\nu,NT}$, let $\mathbb{E}_{0,t}^c$ denote expectation conditional on $\mathcal{F}_{t-1}$ and the current aggregate and common innovations entering the covariates, and let

$$v_{\nu,t}^c = \mathbb{E}_{0,t}^c(\xi_{\nu,t}^2). \quad (5.25)$$

Assumption 6 (Target regularity).

The target satisfies the following conditions.

(i) *Target geometry and nondegenerate variance. The rank and nuclear norm of every nonzero*

component of $\mathbf{D}_\nu$ are uniformly bounded, $\|\mathbf{W}_\nu\|_{\mathbb{H}} > 0$, and

$$\frac{\|\mathbf{D}_\nu\|_{*,1}}{\sqrt{NT}\|\mathbf{W}_\nu\|_{\mathbb{H}}} \longrightarrow 0. \quad (5.26)$$

There exist constants $0 < c_\nu \leq C_\nu < \infty$ such that

$$c_\nu \mathbb{E}_0 \sum_{i=1}^N \sum_{t=1}^T \tilde{\Psi}_{\nu,it}^2 \leq s_{\nu,NT}^2 \leq C_\nu \mathbb{E}_0 \sum_{i=1}^N \sum_{t=1}^T \tilde{\Psi}_{\nu,it}^2. \quad (5.27)$$

If the target selects entry (i, t) of a positive-rank component $\mathbf{M}_0 = \mathbf{U}_M \mathbf{\Sigma}_M \mathbf{V}_M'$, then for some $c_{\text{lev}} > 0$,

$$\|\mathbf{e}'_i \mathbf{U}_M\|_2^2 \geq \frac{c_{\text{lev}}}{N}, \quad \|\mathbf{e}'_t \mathbf{V}_M\|_2^2 \geq \frac{c_{\text{lev}}}{T}. \quad (5.28)$$

(ii) *Gaussian population influence score.* If $\mathcal{D}_{\nu,NT}$ is nonempty, then at every $t \in \mathcal{D}_{\nu,NT}$ and for every i ,

$$\mathbb{E}_0[u_{it}|x_{it}, \mathcal{F}_{t-1}, \text{ current aggregate and common innovations entering the covariates}] = 0. \quad (5.29)$$

This additional centering is needed only for the conditional cross-sectional limit at a locally important period. For the periods outside $\mathcal{D}_{\nu,NT}$,

$$\frac{\sum_{t \notin \mathcal{D}_{\nu,NT}} \mathbb{E}_0[|\xi_{\nu,t}|^{2+\eta/4} | \mathcal{F}_{t-1}]}{s_{\nu,NT}^{2+\eta/4}} \xrightarrow{p} 0. \quad (5.30)$$

There exist deterministic numbers $\tau_{\nu,0}, \dots, \tau_{\nu,K_\nu} \geq 0$ and $\pi_{\nu,1}, \dots, \pi_{\nu,K_\nu} > 0$ such that

$$\frac{V_{\nu,k}^{\text{seg}}}{s_{\nu,NT}^2} \xrightarrow{p} \tau_{\nu,k}, \quad \frac{v_{\nu,d_{\nu,k}}^c}{s_{\nu,NT}^2} \xrightarrow{L^1} \pi_{\nu,k}, \quad \sum_{k=0}^{K_\nu} \tau_{\nu,k} + \sum_{k=1}^{K_\nu} \pi_{\nu,k} = 1. \quad (5.31)$$

At every $t \in \mathcal{D}_{\nu,NT}$,

$$\frac{\sum_{i=1}^N \mathbb{E}_{0,t}^c |Z_{\nu,it}|^{2+\eta/4}}{(v_{\nu,t}^c)^{1+\eta/8}} \xrightarrow{p} 0, \quad (5.32)$$

$$\max_{k \leq N} \frac{\mathbb{E}_{0,t}^c \left[\left(\sum_{i: d_N(i,k) \leq r_N} Z_{\nu,it} \right)^2 \right]}{v_{\nu,t}^c} \xrightarrow{p} 0. \quad (5.33)$$

(iii) *Spatial standard error.* The bandwidth satisfies $h_N \rightarrow \infty$ and $h_N < r_N$, and

$$v_N(h_N) b_{NT}^2 \sqrt{\frac{\log(NTJ_{NT})}{m_{NT}}} \longrightarrow 0, \quad \frac{v_N(h_N)^3}{m_{NT}} \longrightarrow 0. \quad (5.34)$$

The covariance omitted beyond the bandwidth is negligible both in aggregate and at the finitely many target periods:

$$\frac{1}{s_{\nu,NT}^2} \sum_{t=1}^T \sum_{\substack{i,k \leq N \\ d_N(i,k) > h_N}} |\mathbb{E}_0[Z_{\nu,it}Z_{\nu,kt} \mid \mathcal{F}_{t-1}]| \xrightarrow{p} 0, \quad (5.35)$$

$$\max_{t \in \mathcal{D}_{\nu,NT}} \frac{\sum_{\substack{i,k \leq N \\ d_N(i,k) > h_N}} |\mathbb{E}_{0,t}^c[Z_{\nu,it}Z_{\nu,kt}]|}{v_{\nu,t}^c} \xrightarrow{p} 0. \quad (5.36)$$

An expression maximized over an empty $\mathcal{D}_{\nu,NT}$ is interpreted as zero. Finally,

$$\frac{v_N(h_N)^3}{s_{\nu,NT}^4} \sum_{t=1}^T \sum_{i=1}^N \mathbb{E}_0[Z_{\nu,it}^4 \mid \mathcal{F}_{t-1}] \xrightarrow{p} 0. \quad (5.37)$$

For a fixed finite target vector, these conditions hold uniformly over variance-normalized linear combinations, and the limiting correlation matrix is positive definite.

Part (i) says that the target is identified and has a nondegenerate influence variance. For an entry, (5.28) requires the target unit and target time to load nontrivially on the corresponding low-rank component. Part (ii) permits finitely many target periods to remain important: the remaining periods satisfy a martingale central limit theorem, while each important period satisfies a conditional spatial central limit theorem. Part (iii) requires the reported spatial bandwidth to capture the relevant cross-sectional covariance and keeps the quadratic variance estimator from becoming too noisy. All three parts concern population objects or researcher-controlled sequences. Estimated-weight stability and the second-order debiasing remainder are derived in Appendix ??–??.

Population coercivity and the Riesz equation imply

$$\mathbb{E}_0 \sum_{i,t} \tilde{\Psi}_{\nu,it}^2 \asymp \|\mathbf{W}_{\nu}\|_{\mathbb{H}}^2. \quad (5.38)$$

Consequently, Assumption 6(i) gives $s_{\nu,NT} \asymp \|\mathbf{W}_{\nu}\|_{\mathbb{H}}$ and makes (5.26) sufficient for the second-order loading-factor product. For a regular fixed entry, Assumption 4 and (5.28) give

$$c_{\nu}(N^{-1} + T^{-1}) \leq s_{\nu,NT}^2 \leq C_{\nu}(N^{-1} + T^{-1}). \quad (5.39)$$